

2020 青物

1. 1) 无f, 则滑轮不会转动, 不存在角速度

$$T_1 - mg = ma \Rightarrow a = g$$

$$T_1 = 2mg$$

$$2mg = mg + f$$

$$f = \mu N \Rightarrow f = \mu 3mg = mg \Rightarrow \mu = \frac{1}{3}$$

$$N = 2mg + mg$$

$$2) \quad f = \mu N \quad 2mg - T_2 = 2ma$$

$$T_1 - mg = ma$$

$$(T_2 - T_1) \cdot R = \frac{1}{2} \cdot 2mR^2 \cdot \beta \Rightarrow a = \frac{1}{4}g$$

$$a = \beta R$$

$$\mu 3mgR \cdot \frac{1}{2} \cdot 2mR^2 \beta = mR \cdot a = mR \frac{1}{4}g \Rightarrow \mu > \frac{1}{12}$$

$$3) \quad \mu = \frac{1}{12}$$

$$2mg - T_2 = 2ma$$

$$T_1 - mg = ma \Rightarrow a = \frac{7}{24}g$$

$$T_2 = T_1 + \frac{\mu 3mgR}{2}$$

$$2 \quad y = A \cos(\omega(vt - \frac{x}{\lambda}))$$

$$1) \quad \omega = \frac{2\pi}{T} \quad y_p = A \cos(2\pi vt - \frac{3}{2}\pi)$$

反射波, 存在半波损失, 即有

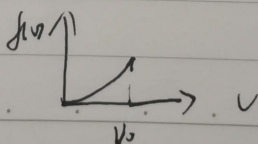
$$y_p' = A \cos 2\pi vt = A \cos \omega t; \quad y_p$$

$$y_p' = A \cos 2\pi \omega (vt + \frac{x}{\lambda})$$

$$12) \quad y = 2A \cos \frac{2\pi x}{\lambda} \cos 2\pi \omega t$$

$$x: \frac{1}{8}\lambda \sim \lambda \quad y: \frac{1}{3}A \cos 2\pi \omega t$$

$$3) \quad f(v) = Av^2$$



$$1.7) \int f(v) dv = 1 \Rightarrow \frac{1}{3} A v_0^3 = 1 \Rightarrow A = \frac{3}{v_0^3}$$

$$1.8) v_p = v_0$$

$$\bar{v} = \frac{\int f(v) v dv}{\int f(v) dv} = \frac{\int A v^4 dv}{1} = \frac{3}{v_0^3} \cdot \frac{1}{5} v_0^5 = \frac{3}{5} v_0$$

$$\sqrt{\bar{v}} = \left(\frac{\int f(v) v^2 dv}{\int f(v) dv} \right)^{1/2} = \left(\frac{3}{v_0^3} \cdot \frac{1}{7} v_0^7 \right)^{1/2} = \sqrt{\frac{3}{7}} v_0$$

$$1.9) \sqrt{\bar{v}} = \left(\frac{\int_0^{v_0} \frac{3}{v_0^3} v^4 dv}{\int_0^{v_0} \frac{3}{v_0^3} v^3 dv} \right)^{1/2} = \frac{1}{2} \sqrt{\frac{3}{5}} v_0$$

$$4.1) \bar{v} = \frac{d\phi}{dt} = \frac{\phi}{t} = \frac{\rho \omega r^2 \sin \omega t}{2\pi} = \frac{1}{2} \rho \omega r^2 = \frac{1}{2} \rho \omega r^2 \beta c$$

$$1.1) \int \rho(r, t) dl = m_0 l$$

$$B \cdot 2\pi r = m_0 \cdot \frac{1}{2} \rho \omega r^2 \beta c \Rightarrow B = \frac{m_0}{4\pi} \rho \omega r \beta c$$

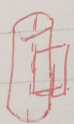
$$1.3) \int E dl = - \int \frac{\partial \phi}{\partial t} ds$$

$$E \cdot 2\pi r = - \frac{\partial \phi}{\partial t} 2\pi r^2$$

$$E = - \frac{m_0}{4\pi} \rho \omega r \beta c$$

$$1.4) dI = \rho \omega r \beta c dr$$

$$B \cdot dl = m_0 dI \Rightarrow B \cdot l = \int_0^R \rho \omega r \beta c \cdot r dr \Rightarrow B = \frac{m_0}{2} \rho \omega \beta c (R^2 - r^2)$$



$$1.5) \int E dl = - \int \frac{\partial \phi}{\partial t} \cdot ds$$

$$E \cdot 2\pi r = - \int \frac{m_0 \rho \omega \beta c}{2} (R^2 - r^2) ds = \frac{m_0 \rho \omega \beta c}{8}$$

$$\Rightarrow E = \frac{1}{8} m_0 \rho \omega \beta c (2R^2 - r^2)$$

$$5.1) d \sin \theta = k \lambda, \quad d = \frac{1 \text{ nm}}{500} = 2000 \text{ nm}, \quad \lambda = 1000 \text{ nm}$$

$$\sin \theta \cdot d = k \lambda < d \Rightarrow k < \frac{d}{\lambda}, \quad \text{能看到的级数}$$

$$1.1) d \sin \theta \neq a \cdot \sin \theta = k \lambda$$

$$d \sin \theta = k \lambda + a \sin \theta < d \Rightarrow k < \frac{d \sin \theta + a}{\lambda}$$

$$6. \text{ (1)} \quad E = \frac{Q(t)}{4\pi\epsilon_0 r^2} \quad j = GE = \frac{6Q(t)}{4\pi\epsilon_0 r^2}$$

$$I = j \cdot S = j \cdot 4\pi r^2 = 4\pi r^2 \cdot \frac{6Q(t)}{4\pi\epsilon_0 r^2} = -\frac{dQ(t)}{dt}$$

$$\Rightarrow Q(t) = Q_0 e^{-\frac{t}{RC}}$$

$$(2) \text{ 微分形式 焦耳定律 } w = GE^2$$

$$Q = \int_0^t \int_V w dV dt = \int_0^t \frac{6Q(t)}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b}\right) e^{-\frac{t}{RC}} dt$$

$$= \frac{Q_0^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b}\right) (1 - e^{-\frac{t}{RC}}) \quad (\text{焦耳})$$

$$\text{电容 } C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$\text{静电功 } W = \int_a^b \frac{Q}{r} dr = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b}\right) (e^{-\frac{t}{RC}} - 1)$$

7. (1) 假设上下半球电荷均匀分布:

$$E = \begin{cases} \frac{Q_1}{4\pi\epsilon_1 r^2}, & \text{下} \\ \frac{Q_2}{4\pi\epsilon_2 r^2}, & \text{上} \end{cases}$$

电荷守恒, 即有:

$$\int_{R_1}^{R_2} E_1 dr = \int_{R_1}^{R_2} E_2 dr \Rightarrow \frac{Q_1}{4\pi\epsilon_1} \left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{Q_2}{4\pi\epsilon_2} \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$E_1 = E_2$
(边界)

$$E_1(Q_1 = Q_2, \epsilon_1 + \epsilon_2 = \epsilon)$$

$$\text{即有: } \begin{cases} Q_1 = \frac{\epsilon_2 Q}{1 + \epsilon_2} \\ Q_2 = \frac{\epsilon_1 Q}{1 + \epsilon_1} \end{cases}$$

$$(2) \quad U = \int_{R_1}^{R_2} E_1 dr = \frac{Q_1}{4\pi\epsilon_1} \left(\frac{1}{R_1} - \frac{1}{R_2}\right) = \frac{Q}{4\pi\epsilon_1(1 + \epsilon_1)} \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$(3) \quad U_2 = \frac{Q}{4\pi\epsilon_2 R_2}$$

$$U_1 = U + U_2 = \frac{Q}{4\pi\epsilon_1(1 + \epsilon_1)} \left(\frac{1}{R_1} - \frac{1}{R_2}\right) + \frac{Q}{4\pi\epsilon_2 R_2}$$